

KustoCon

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From Edges to Nodes: Attack Path Discovery in Kusto

Olaf Hartong - FalconForce



KustoCon
Learn | Share | Practice

OLAF HARTONG

Detection Engineer and Security Researcher

- Purple teaming, Threat hunting
- Security MVP

Former documentary photographer
Father of 2 boys
“I like **warm hugs**”

 @olafhartong
 github.com/olafhartong
 olaf@falconforce.nl
 olafhartong.nl / falconforce.nl



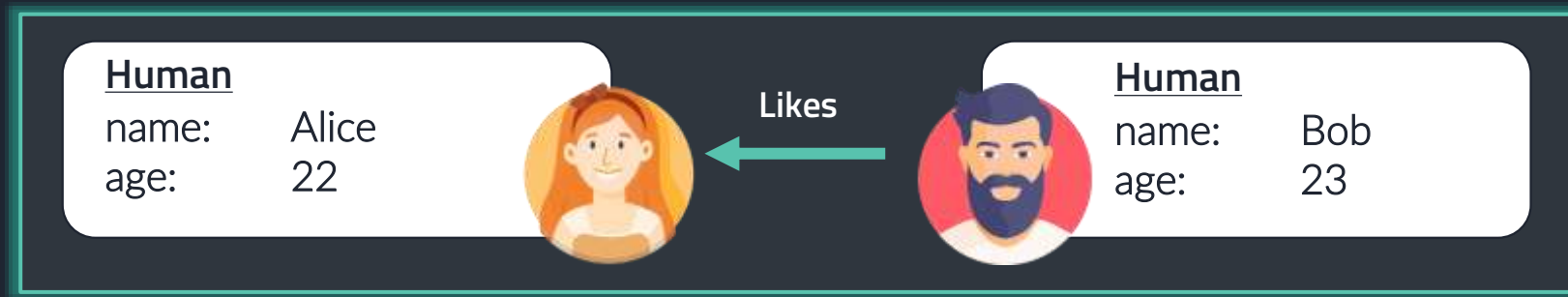
WHAT IS A

GRAPH AND WHY

SHOULD I

CARE?

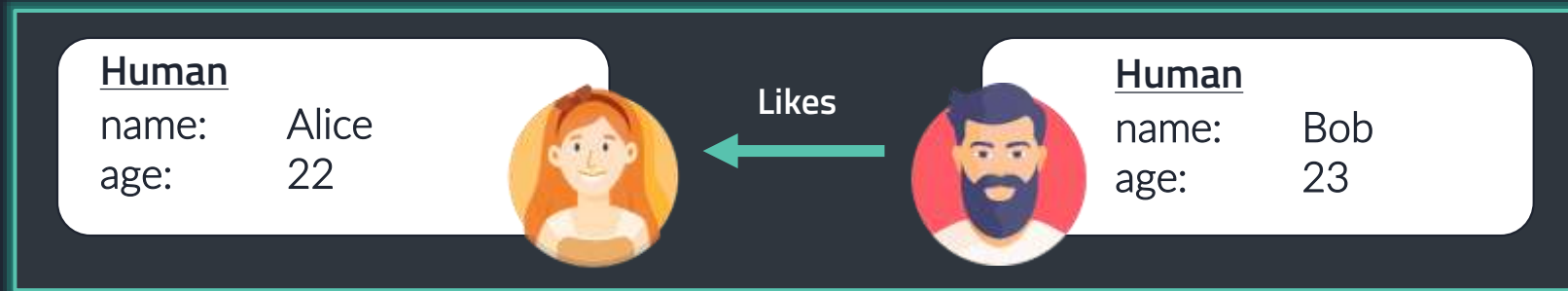
GRAPH CONCEPT & TERMINOLOGY



- Objects are called **Nodes**
- Nodes have a type aka **Label**
- Nodes have **properties**
- Relationships are called an **Edges**

[Human]
[name/age]
[Likes]

GRAPH CONCEPT & TERMINOLOGY



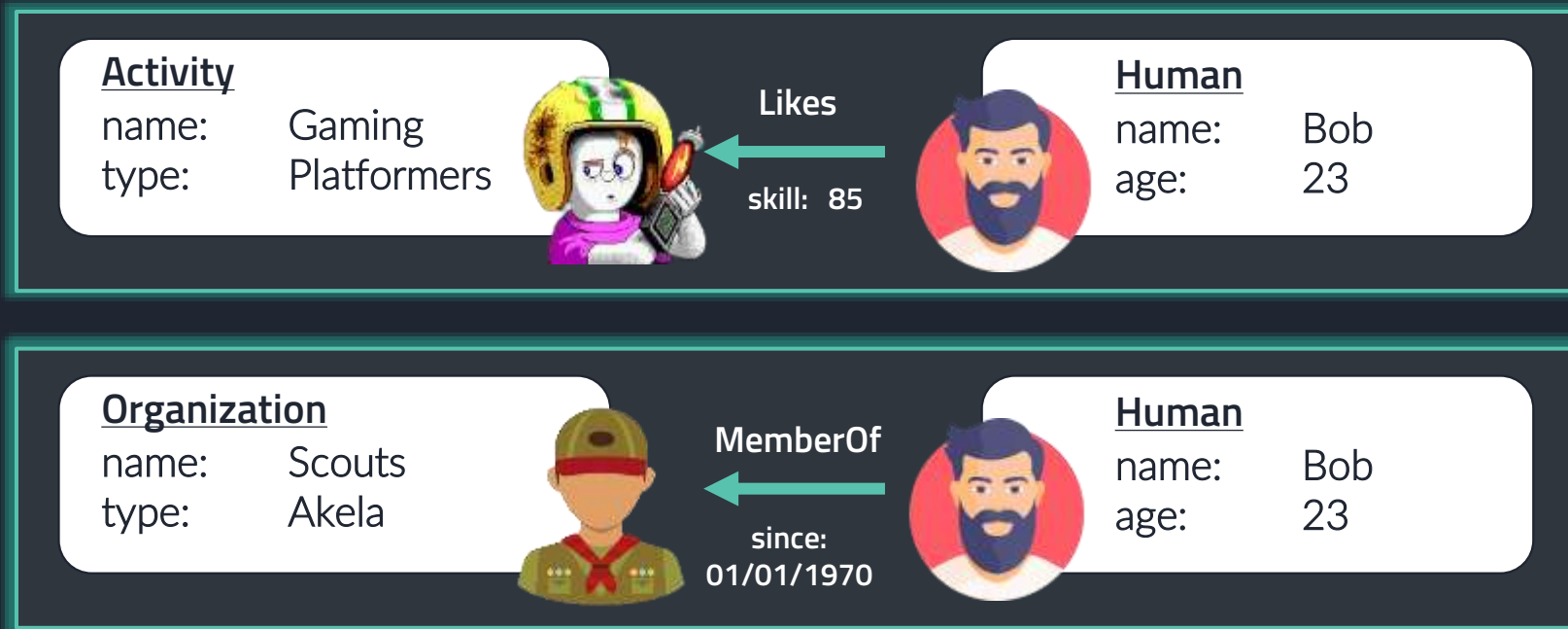
Important:

In our case,
Edges are directional

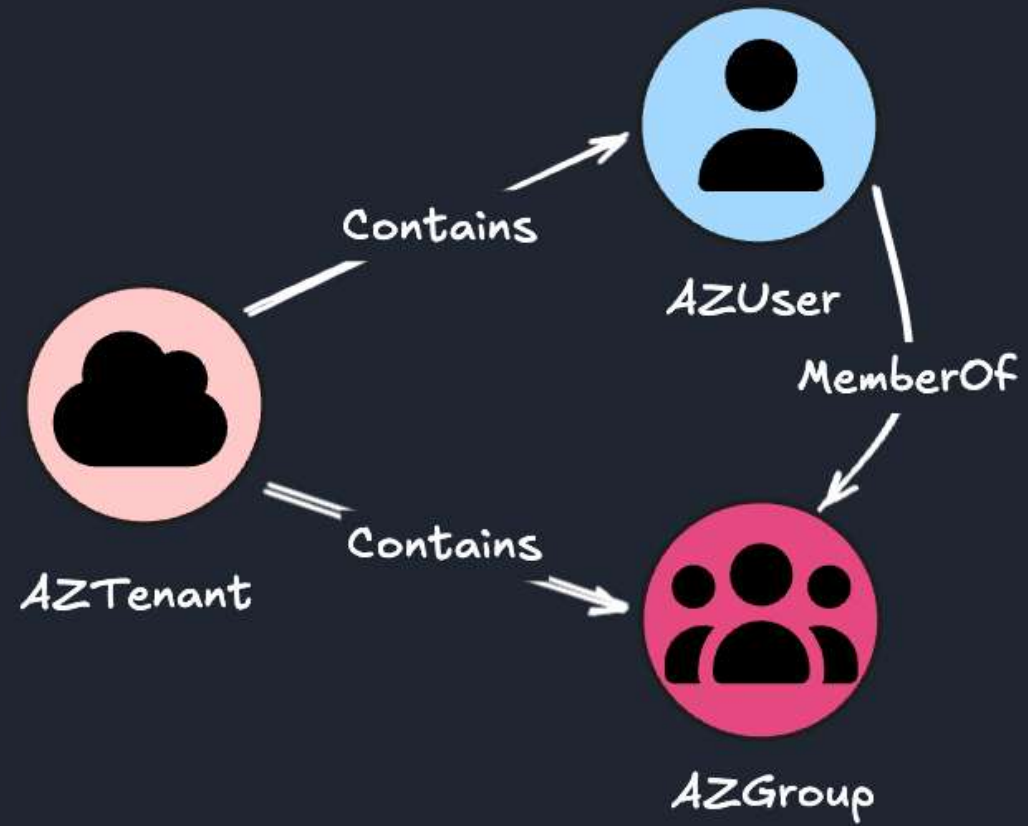
[We need two Edges
for a relationship
to go both
ways...]

GRAPH CONCEPT & TERMINOLOGY

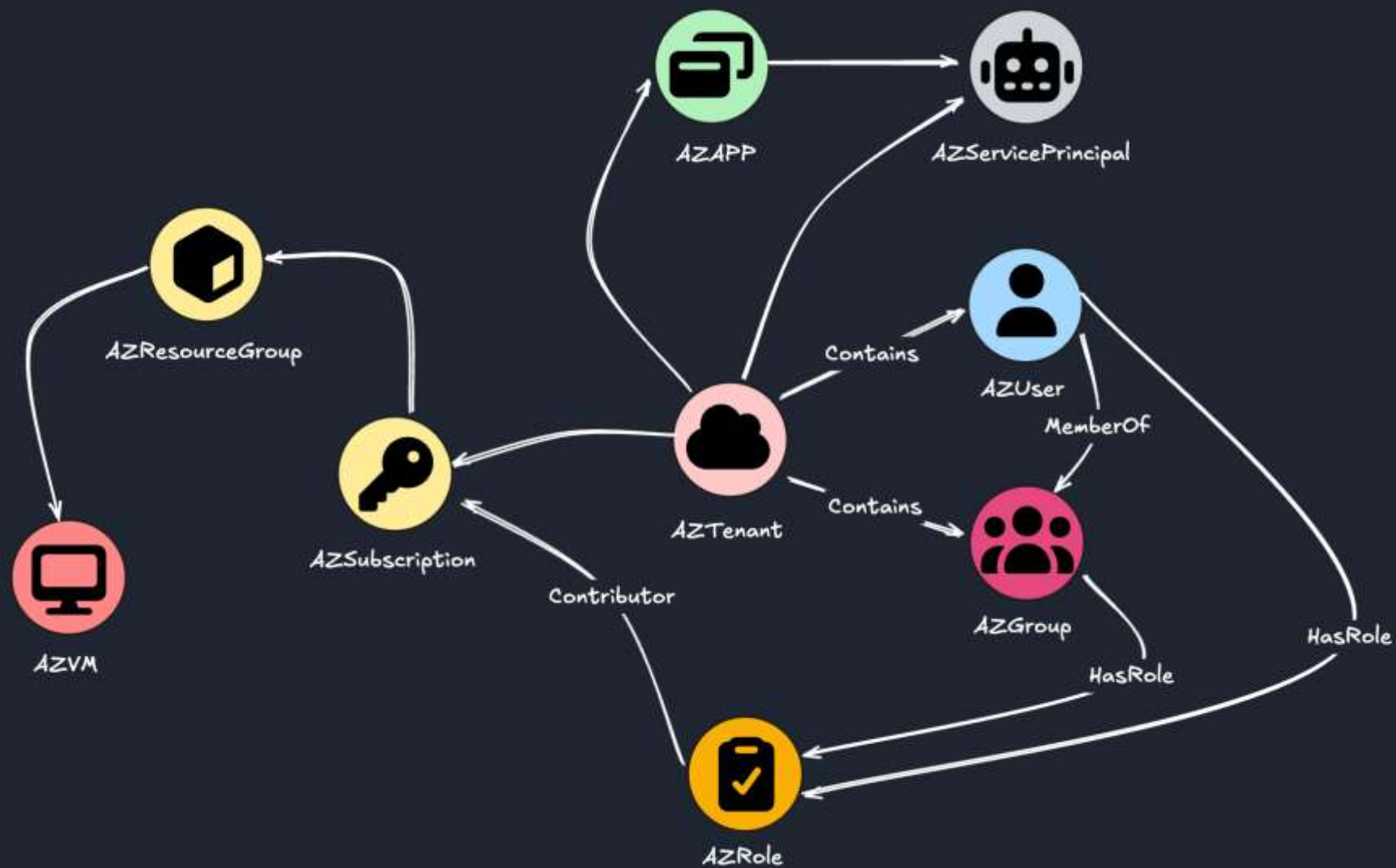
Edges can exist between any node, and like nodes they can have properties too



SIMPLIFIED AZURE EXAMPLE



SIMPLIFIED AZURE EXAMPLE



GRAPH OPERATORS

make-graph

// create a graph in memory from tabular data

graph-match

// query the graph with a specific KQL pattern notation, returns tabular data

graph-shortest-paths (preview)

// query the graph, returns tabular data of 1 shortest path.

// this can be configured to show all shortest paths

graph-to-table

// convert a graph into tabular data

GRAPH TYPES

Dynamic graphs / Mixed mode graphs

1

Mix a graph with known nodes and edges and search results

2

Requires a strong design and prefiltering of data

3

Very flexible, but potential performance risk

Prebuilt graphs (snapshots)*

1

Create nodes and edges for all relevant data periodically

2

Requires some scheduling mechanism to remain to be up-to-date

3

Less flexible, but much more performant. This can be even faster with snapshots

** Depending on availability in the platform*

QUERYING A KQL GRAPH

Directly as shown before with the make-graph or a function for the KQL make-graph

```
...
Edges
  | make-graph sourceId --> targetId with Nodes on id

// or when defined as a function
Attack-graph
```

Calling a graph model

```
graph('AzureHound')
```


GRAPH MODELS CAN BE MORE POWERFUL

Calling a graph model

```
graph('AzureHound')
```

When running this, it will run the last created snapshot.

If there are no snapshots, it will run on the live data

You can force this by adding a parameter

```
graph('AzureHound', true) //transient mode, to get live data
```

Or even specify which snapshot you want to run

```
graph('AzureHound', `warmHugs`) //query named snapshot
```

GRAPH SNAPSHOTS

- Point-in-time graph - Represents the graph state at creation time *
- Persistent: It will be stored in the database until explicitly dropped
- Direct querying: Enables queries without rebuilding the graph

Snapshots are very useful and much faster BUT:

Be aware you need to keep them **up-to-date** and think about **lifecycle management**, they will incur storage and therefore cost.

** Snapshots can be epic for comparisons over time*

QUERYING A KQL GRAPH

arbitrary names

```
|graph-match (theuser)-[HasPermission*3..5]->(azureResource)
```

path length

QUERYING A KQL GRAPH

```
|graph-match (theuser)-[HasPermission*3..5]->(azureResource)  
  where theuser.NodeName contains "Rogier" and  
  azureResource.NodeLabel has_any ("virtualmachines",  
  "resourcegroups", "subscriptions")
```

No pipe within the match

Filters need to be more extensive since the node / edge names are not related to the objects

QUERYING A KQL GRAPH

```
|graph-match (theuser)-[HasPermission*3..5]->(azureResource)
  where theuser.NodeName contains "Rogier" and
  azureResource.NodeLabel has_any ("virtualmachines",
  "resourcegroups", "subscriptions")
  project theuser=theuser.NodeName, Path =
  HasPermission.EdgeLabel, TheResource=azureResource.NodeName,
  TheLabel=azureResource.NodeLabel
```

You need to exactly specify what to return (project)

BASIC KQL EXAMPLE

```
let employees = datatable(name:string, age:long)
[
  "Alice", 32,
  "Bob", 31,
  "Eve", 27,
  "Joe", 29,
  "Chris", 45,
  "Alex", 35,
  "Ben", 23,
  "Richard", 39,
];
let reports = datatable(employee:string, manager:string)
[
  "Bob", "Alice",
  "Chris", "Alice",
  "Eve", "Bob",
  "Ben", "Chris",
  "Joe", "Alice",
  "Richard", "Bob"
];
reports
| make-graph employee --> manager with employees on name
| graph-match (alice)<-[reports*1..5]-(employee)
  where alice.name == "Alice" and employee.age < 30
  project employee = employee.name, age = employee.age, reportingPath = map(reports, manager)
```

BASIC KQL EXAMPLE

```
let employees = datatable(name:string, age:long)
[
  "Alice", 32,
  "Bob", 31,
  "Eve", 27,
  "Joe", 29,
  "Chris", 45,
  "Alex", 35,
  "Ben", 23,
  "Richard", 39,
];
let reports = datatable(employee:string, manager:string)
[
  "Bob", "Alice",
  "Chris", "Alice",
  "Eve", "Bob",
  "Ben", "Chris",
  "Joe", "Alice",
  "Richard", "Bob"
];
reports
| make-graph employee --> manager with emp
| graph-match (alice)<-[reports*1..5]-(emp
  where alice.name == "Alice" and employee
  project employee = employee.name, age =
```

employee	age	reportingPath
> Joe	29	["Alice"]
> Ben	23	["Alice","Chris"]
> Eve	27	["Alice","Bob"]

PERFORMANCE PATTERNS

It is recommended to use filters to reduce the amount of data being processed and to limit the number of operations being performed on large datasets. This improves performance significantly.

Some basic design principles for your graphs;

- Try to filter as close to the top of the query as possible. Select only relevant data for nodes, edges, and properties before graph creation
- Remove unnecessary columns to minimize memory consumption.
- Consider extracting some data, instead of fully parsing all data.
- Summarize data where appropriate to reduce graph complexity

NOTABLE FACTS

- Transient graphs are constructed in-memory on a single node, limiting them to data that fits in the RAM of that node.
- A maximum of 5,000 graph snapshots are allowed per database (500 for free virtual clusters). Snapshot creation time limit : 1 hour (which is long for most data sets)
- The [graph-shortest-paths](#) operator has limitations on the length and complexity of paths it can find. Patterns must include at least one variable length edge and can't contain multiple sequences.
- Result data can not be greater than 65MB or 500k records
- There was a limit of 10mln nodes+edges, I believe this to be much higher now

WHAT TO

GET FOR HIM?

GRAPH OPTIONS

Exposure graph

Phishing attempts

Process execution
trees

Incident correlation

Entra + Azure

Patient 0
analysis

Attack paths

Vulnerabilities

Network
communication

INCIDENT GRAPHS

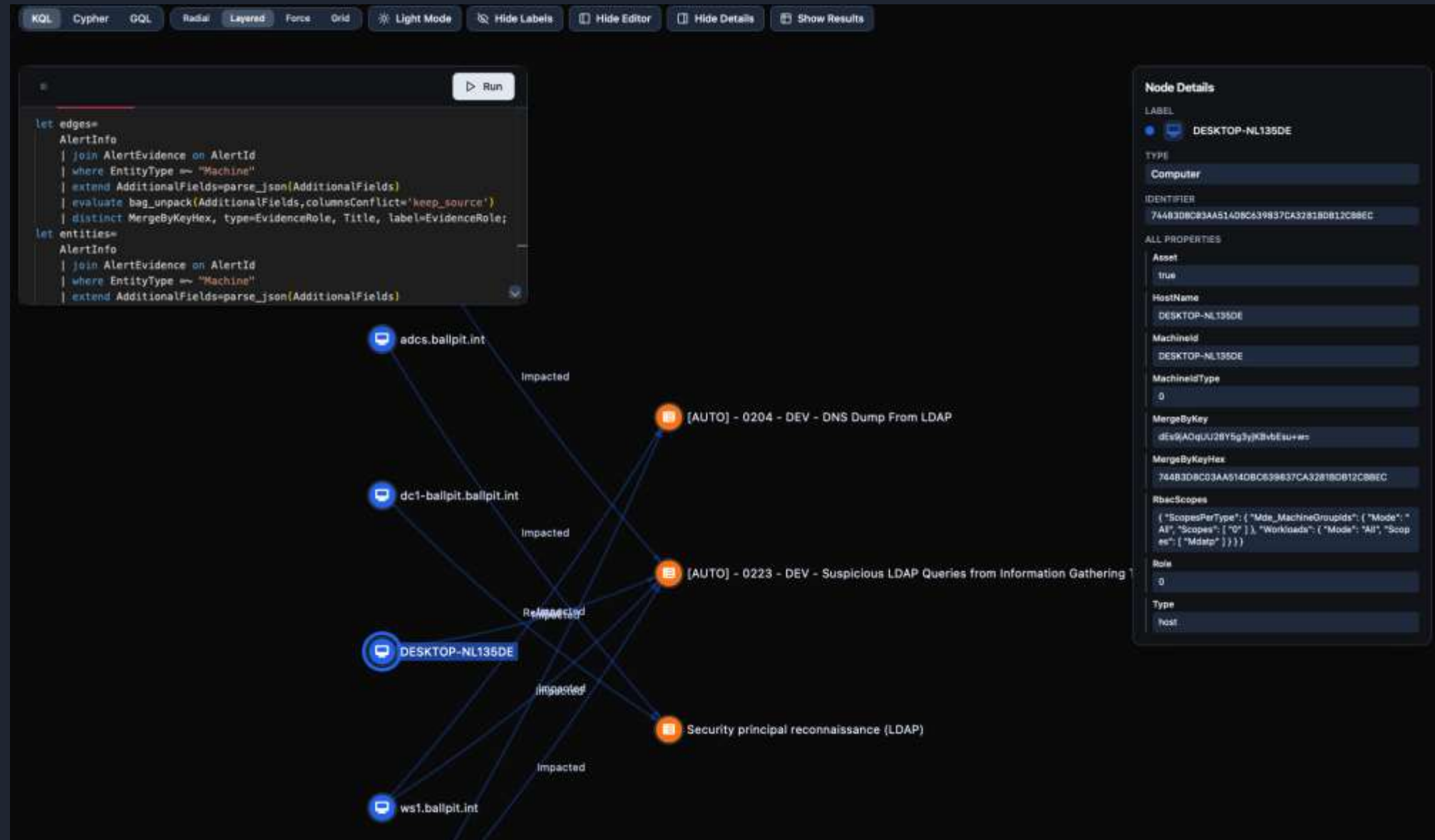


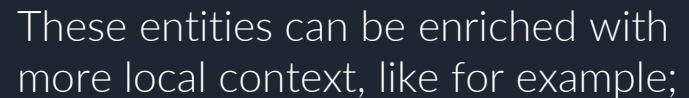
ALERT GRAPH EXAMPLE

Graphing machines impacted by the same alerts

```
let edges=
  AlertInfo
  | join AlertEvidence on AlertId
  | where EntityType =~ "Machine"
  | extend AdditionalFields=parse_json(AdditionalFields)
  | evaluate bag_unpack(AdditionalFields,columnsConflict='keep_source')
  | distinct MergeByKeyHex, type=EvidenceRole, Title, label=EvidenceRole;
let entities=
  AlertInfo
  | join AlertEvidence on AlertId
  | where EntityType =~ "Machine"
  | extend AdditionalFields=parse_json(AdditionalFields)
  | extend MergeByKeyHex=AdditionalFields.MergeByKeyHex
  | distinct id=toString(MergeByKeyHex), properties=toString(AdditionalFields), label=toString(DeviceName);
let alerts=
  AlertInfo
  | join AlertEvidence on AlertId
  | where EntityType =~ "Machine"
  | extend AdditionalFields=parse_json(AdditionalFields)
  | extend MergeByKeyHex=AdditionalFields.MergeByKeyHex
  | distinct id=toString(Title), label=Title, Category, Severity, ServiceSource, DetectionSource, AttackTechniques;
let nodes=union entities,alerts;
edges
| make-graph MergeByKeyHex --> Title with nodes on id
| graph-match (n1)-[e]->(n2)-[r]-() project from=n1, via=e, to=n2
```


RELATED ALERT GRAPH EXAMPLE





- Much, much more



EXPOSURE
GRAPH

EXPOSURE GRAPH

- Security related findings and recommendations
- Built and maintained by Microsoft
- Completeness of the information depends on your **licenses**
- Only shows what MS determines as risks, not all misconfigurations
- Querying this data seems pretty slow

```

44 ExposureGraphEdges
45 | where SourceNodeLabel !~ "Cve"
46 | make-graph SourceNodeId --> TargetNodeId with ExposureGraphNodes
47 on NodeId
48 | graph-match (SourceNode)-[edges]->(TargetNode)
49   where TargetNode.NodeName contains "Eileen" or SourceNode.NodeName contains "Eileen"
50   project IncomingNodeLabels = SourceNode.NodeLabel, SourceNode.NodeName, EdgeLabel=edges.EdgeLabel, TargetNodeName= TargetNode.NodeName, TargetNodeLabel=TargetNode.NodeLabelz

```

Getting started **Results** Query history

↓ Export ▾  Show empty columns

5 items

 Search

 01:29.459  Low 

Filters:  Add filter

☐ IncomingNodeLabels	SourceNode_NodeName	EdgeLabel	TargetNodeName	TargetNodeLabel
☐ > aadSecurityRecommendation	SigninRiskPolicy	affecting	Eileen Sideways	user
☐ > aadSecurityRecommendation	UserRiskPolicy	affecting	Eileen Sideways	user
☐ > aadSecurityRecommendation	BlockLegacyAuthentication	affecting	Eileen Sideways	user
☐ > aadSecurityRecommendation	MfaRegistrationV2	affecting	Eileen Sideways	user
☐ > user	Eileen Sideways	member of	All Users	group

Layout Algorithm

Force-Directed

☒ Show Node Labels

☒ Show Edge Labels

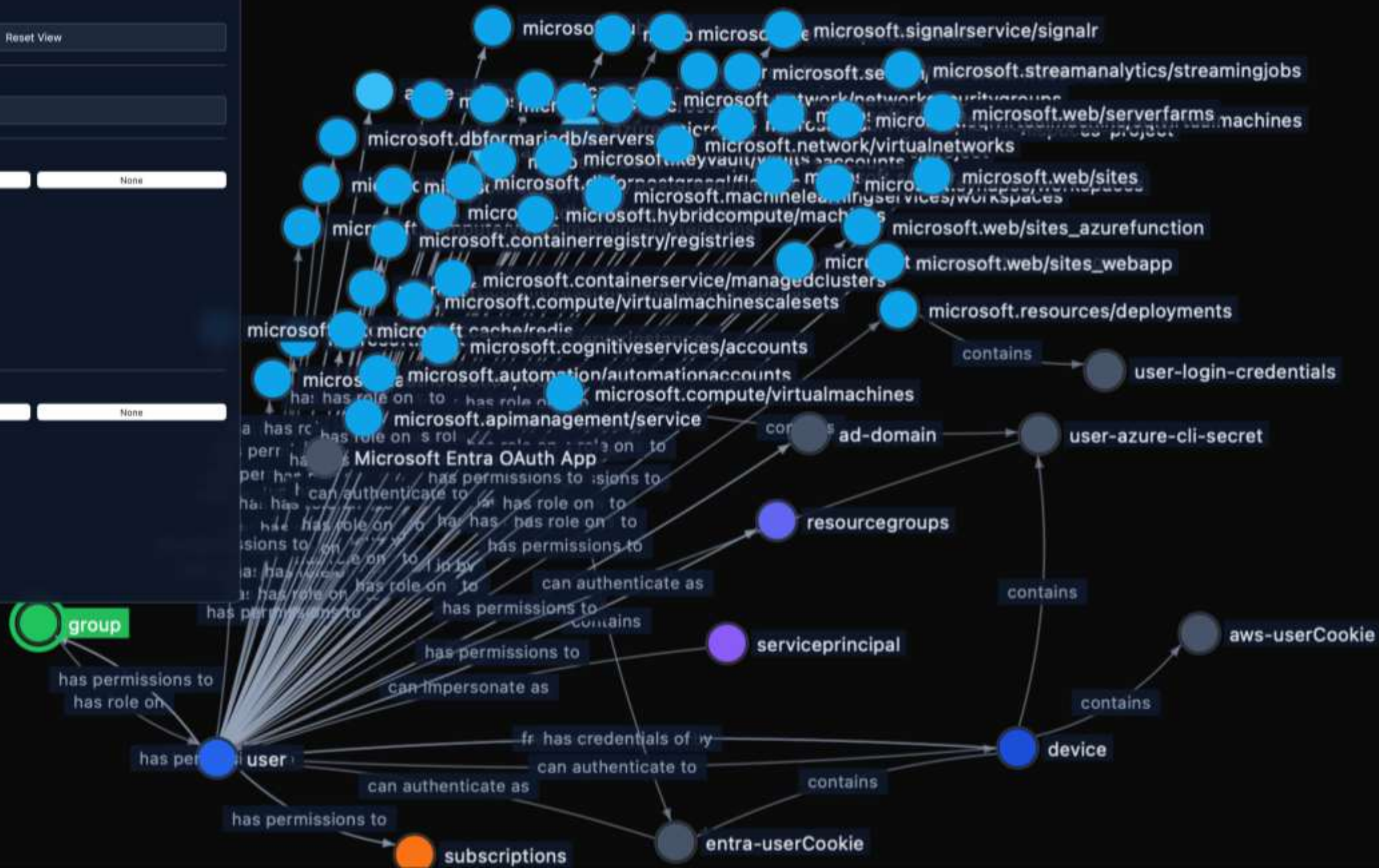
Reset View

☐ All ☐ None

- ### Edge Types (18/18)

All None

- ✖ affecting
- ✖ can authenticate as
- ✖ can authenticate to
- ✖ can impersonate as
- ✖ contains
- ✖ defined in
- ✖ defines
- ✖ frequently logged in by
- ✖ has credentials of
- ✖ has permissions to



Visible Nodes: 62 / 154
Visible Edges: 117 / 667
Node Types: 154

Layout Algorithm

Force-Directed

☒ Show Node Labels

☒ Show Edge Labels

Reset View

Search

managedidentity

Node Types (154/154)

A11

None

☒ BlobContainer

Cve

 FileShare

☐ IP address

K&S-Conta

- ✓ K8s-daemons

- ✓ K8s-deployment

☒ K8s-ingress

KBS—narrowed

of the same magnitude.

Edge Types (18/18)

A11

None

☒ affecting

 can authenticate as

 can authenticate to

 can impersonate us.

☒ contains

 defined in

 defines

☒ frequently

 has credentials of

► [View our complete catalog](#) ►

Visible Nodes: 54 / 154

Visible Edges: 107 / 657

Node Types: 154

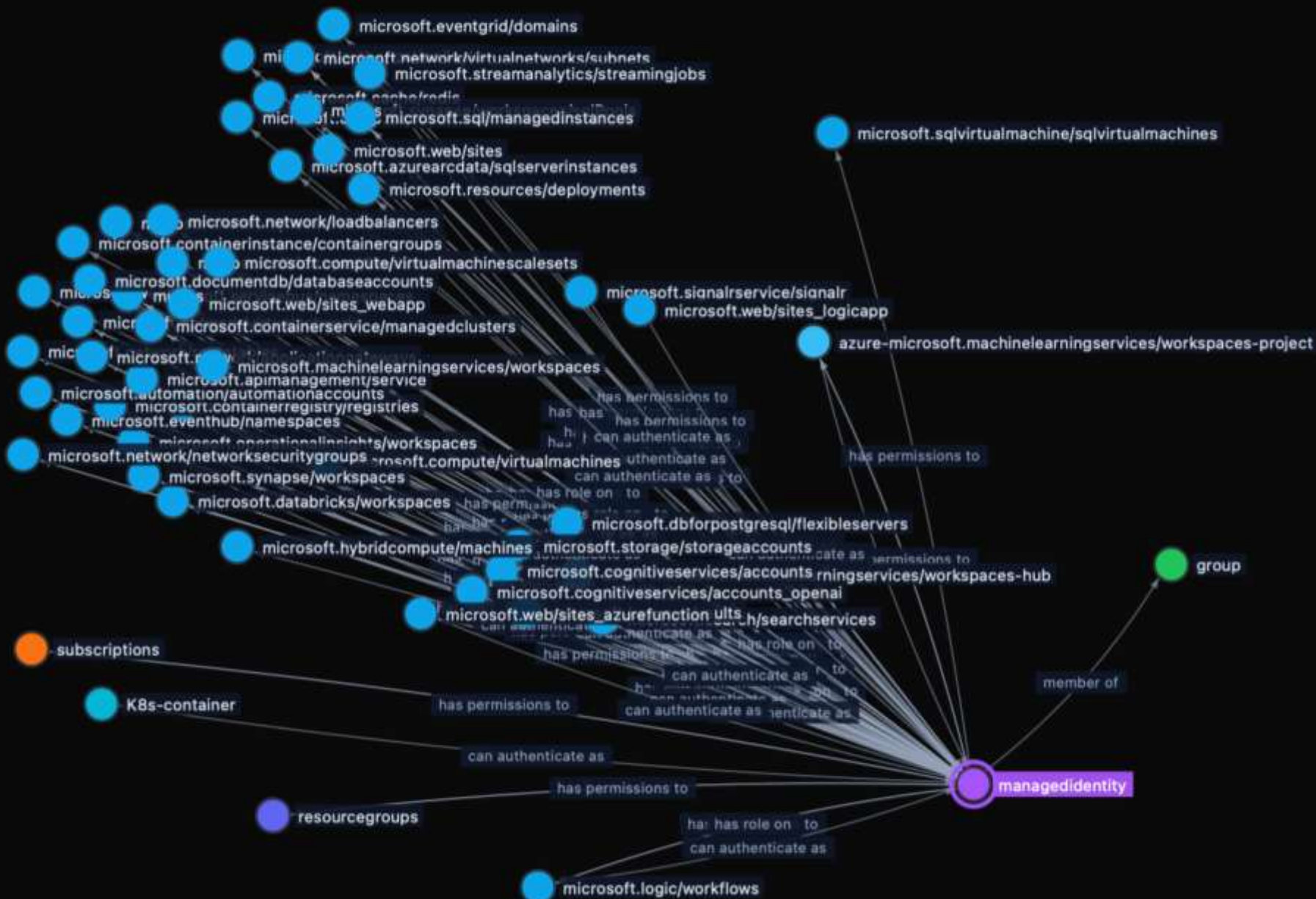
managedidentity

Type: managed_identity

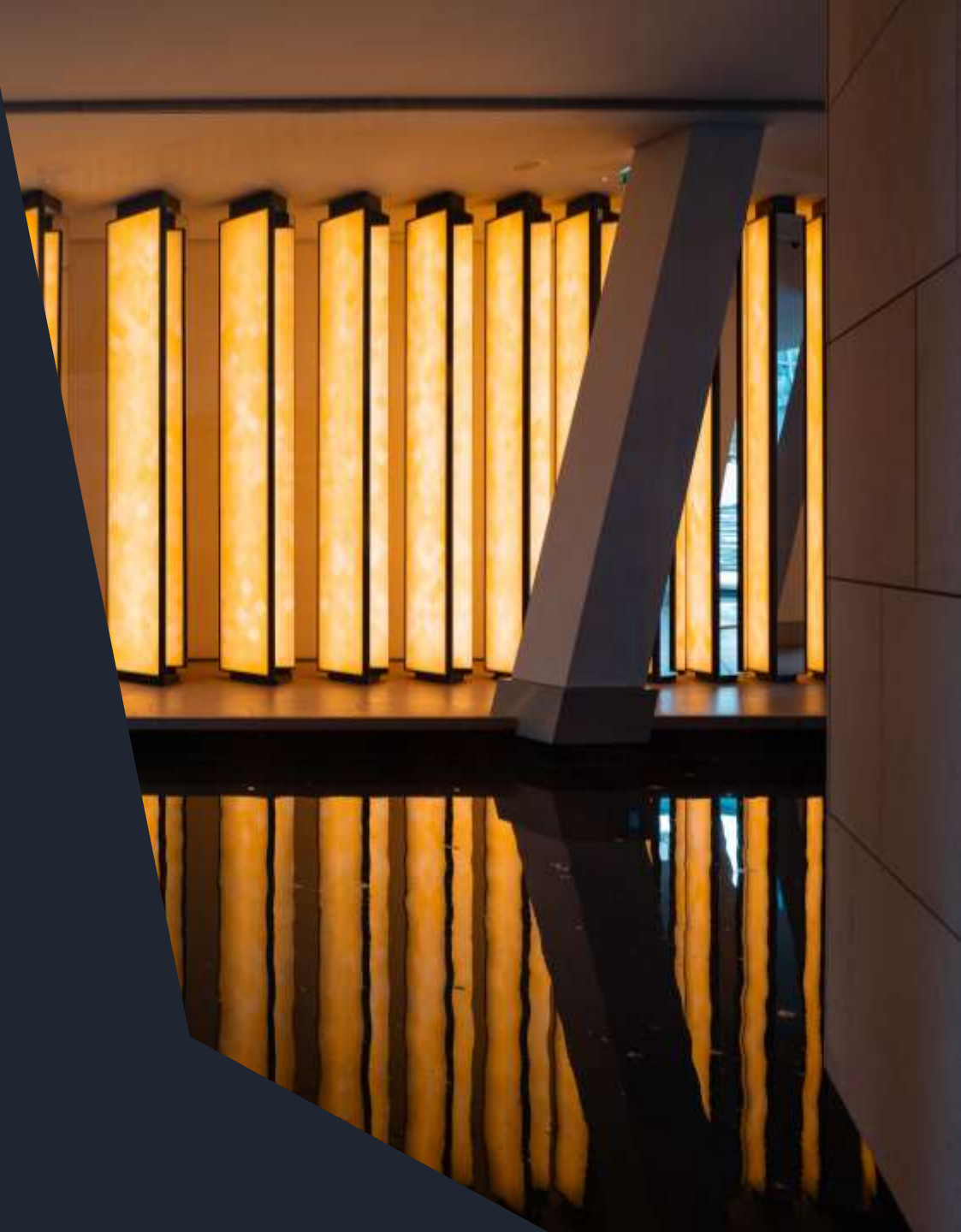
Connections: 107

incoming: 28

Outgoing: 81



SENTINEL DATA LAKE



SENTINEL DATALAKE

Once enabled it will populate tables with EntraID and Azure Resource Manager information

With some effort, can be turned into a graph

Built and maintained by Microsoft

Data is updated periodically, no control over the frequency, for example, taken on the 28th, the data is almost 2 days old

_SnapshotTime	2025-10-26T22:00:00Z
_ReceivedTime	2025-10-27T08:15:00Z
TimeGenerated	2025-10-26T22:00:00Z

Assets

- >  ARGAuthorizationResources
- >  ARGResourceContainers
- >  ARGResources
- >  EntraApplications
- >  EntraGroupMemberships
- >  EntraGroups
- >  EntraMembers
- >  EntraOrganizations
- >  EntraServicePrincipals
- >  EntraUsers

[Run query](#)

Last 24 hours

[Create Job](#)

```
1 let Nodes=union EntraUsers,EntraServicePrincipals,EntraApplications,EntraGroups, EntraMembers;
2 EntraMembers
3 | extend edge_type=case(sourceType == "group", "memberOf", "hasRole")
4 | make-graph sourceId --> targetId with Nodes on id
5 | graph-match (n1)-[e]->(n2) where n2.userPrincipalName contains "Olaf@" project n1.id, n1.displayName,e.edge_type,n2.displayName
6 | distinct *
7
```

[Getting started](#)[Results](#)[Query history](#)[Export](#)

11 items

3.14 sec

[Search](#)[Customize columns](#)

n1_id	n1_displayName	e_edge_type	n2_displayName
> d9e65920-31d4-44a7...	SentryDevTeam	memberOf	Olaf Hartong
> c8e72849-3b50-472c...	falconforceballpit	memberOf	Olaf Hartong
> f85bd02f-9177-4bca-...	0xff	memberOf	Olaf Hartong
> 24ac8607-7d59-4b13...	DynamicGroup	memberOf	Olaf Hartong
> 84163c0e-dc96-403f...		hasRole	Olaf Hartong
> 3aec2ad0-ae08-475a...		hasRole	Olaf Hartong
> 5fbb23de-9286-44f5...		hasRole	Olaf Hartong
> a1aef518-8c8e-4243...		hasRole	Olaf Hartong
> 94ba6eda-4f23-4d44...		hasRole	Olaf Hartong
> 3af23415-ac9b-4467...		hasRole	Olaf Hartong
> 19b3d281-c5dc-45ae...		hasRole	Olaf Hartong

SOME PROS AND CONS

- 'free' and no maintenance
- Graph support, including GQL and maybe OpenCypher soon?
- After graphing useful for enrichment
- Far from complete (yet)
 - No role details
 - No api permissions
 - No permissions to resources not based on a role
 - Slow updates
 - No control over the updates
 - Querying is not very fast
 - No official API (yet)



ATTACK PATHS

ATTACK PATHS



Platform Services & Trademarks

Attack paths



Platform Products Solutions Why Tenable Resources

What is Attack Path Management?

Attack Paths are nothing new. Identifying them are particularly well-studied. Attackers create direct and indirect connections between assets to move laterally across a network.

Estimated read time: 30 min

What are attack paths?

Directory services: Active Directory

Why Active Directory is the adversary's favorite target

Comparing Attack Path Management and vulnerability management

Why attack paths are challenging

Why current options to address attack paths fall short

What is Attack Path Management and how it works

Attack Path Management benefits

Cybersecurity Guide / Attack Path Management / Attack Path Management

UNDERSTANDING ATTACK PATH MANAGEMENT: THE FUNDAMENTALS

Is APM Part of Your Overall Cybersecurity Strategy?

Attack path management (APM) is a process your organization can use to identify and remediate security weaknesses as seen through the eyes of an attacker. By understanding attack paths within your enterprise, you can build stronger security defenses and enable teams to quickly cut off these attack pathways and shut down attacks before they move deeper into your systems and network. These attack pathways include on-prem assets and systems, as well as within your cloud environment.

In this knowledge base about attack path management, we take a close look at all about, how it can help mature your overall cybersecurity practices, and how it can help you have more comprehensive insight into your vulnerabilities and security on an ongoing basis.

Here are few highlights of what you'll discover:

Know Your Attack Paths to Bolster Your Defenses

Organizations of all sizes are susceptible to a cyber-attack. Knowing your attack paths can help you stop an attack in its tracks.

Understanding Attack Path Management

To be better prepared to tackle cyber risks head-on, it helps to think like an attacker.



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Attack Path Analysis (APA) Explained



Greg Zemlin

October 11, 2025

6 minute read

Get the Incident Response Template

Watch 5-min demo



Key takeaways about attack path analysis:

- Attack path analysis (APA) maps the sequence of exploitable weaknesses an attacker could use to breach a system and reach critical assets.
- It differs from an attack vector, which is the initial entry point, by detailing the entire chain of lateral movement post-breach.
- The core benefit of APA is proactive risk management, allowing teams to prioritize vulnerabilities that are part of a viable attack path rather than treating all alerts equally. For example, in 2019, an attacker **exploited a vulnerability** in a cloud firewall configuration to access the data of over 100 million Capital One customers, highlighting how a single weakness can create a major breach.
- Effective APA relies on a graph-based model to visualize complex relationships between resources, permissions, and vulnerabilities across a cloud environment.

Table of contents

What is attack path analysis?

What are attack vectors and attack paths?

How does attack path analysis work?

Benefits of attack path analysis

Analyzing and prioritizing attack paths with Wiz

Super-charge your cloud security with Wiz attack path analysis

See Wiz in Action

Frequently asked questions about attack path analysis



WHY

Let's assume this cloud user in our tenant, named Eileen is compromised.

There is no way to see what the potential impact may be, since we have no 'easy' way to see what she has access to in the Azure portal

The screenshot shows the Azure portal interface for a user named Eileen Sideways. The left sidebar contains navigation links: Overview, Audit logs, Sign-in logs, Diagnose and solve problems, Custom security attributes, Assigned roles, Administrative units, Groups, Applications, Licenses, Devices, Azure role assignments, Authentication methods, and New support request. The main content area displays the user's profile with a warning banner: "This user is a member of a restricted management administrative unit. Management rights are limited to administrators scoped on that administrative unit." Below the banner are tabs for Overview, Monitoring, and Properties. The Overview tab shows basic info, including a profile picture, name, email, and member status. A table lists user details: User principal name (Eileen.Sideways@falconforce.training), Object ID (41e417c0-4fba-47a3-8511-b4a59e2a14eb), Created date time (Jun 8, 2023, 4:19 PM), User type (Member), Identities (falconfortraining.onmicrosoft.com), and Agent ID. To the right, there are links for Group memberships (1), Applications (0), Assigned roles (0), and Assigned licenses (0). At the bottom, there are three cards: Account status (Enabled), Sign-ins (Last interactive sign-in: Mar 19, 2025, 4:11 PM), and B2B invitation (Convert to external user).

Home > FalconForce Training | Users > Users >

Eileen Sideways
User

Search

Edit properties Delete Refresh Reset password Revoke sessions Manage view Got feedback?

This user is a member of a restricted management administrative unit. Management rights are limited to administrators scoped on that administrative unit.

Overview Monitoring Properties

Basic info

 **Eileen Sideways**
Eileen.Sideways@falconforce.training
Member

User principal name Eileen.Sideways@falconforce.training

Object ID 41e417c0-4fba-47a3-8511-b4a59e2a14eb

Created date time Jun 8, 2023, 4:19 PM

User type Member

Identities falconfortraining.onmicrosoft.com

Agent ID

Group memberships 1

Applications 0

Assigned roles 0

Assigned licenses 0

My Feed

Account status
Enabled
Edit

Sign-ins
Last interactive sign-in: Mar 19, 2025, 4:11 PM
Last non-interactive sign-in: null
See all sign-ins

B2B invitation
Convert to external user

Let
our
cor

The
pot
we
wh
Azu

```
MATCH p=(u:AZUser)-[*1..4]->(z:AZRole) WHERE  
CONTAINS "Eileen" AND z.properties.displayName  
CONTAINS "Administrator" RETURN p
```

Home > FalconForce Training > App registrations > Weather prediction app (Demo)

Weather prediction app (Demo) | Owners

Search + Add owners Remove owners Got feedback?

The users listed here can view and edit this application registration. Additionally, any user (may not be listed here) with administrative privileges to manage any application (e.g., Global Administrator, Cloud App Administrator etc.) can view and edit the application registrations. Currently, only individual users are supported as owners of applications. Assignment of groups as owners is not yet supported. If the user setting "Restrict access to Microsoft Entra ID administration portal" is set to Yes, non-admin users will not be able to use the Azure portal to manage the applications they own. [Learn more](#)

Name	Email	User name	Job Title	Type
☐  Eileen Sideways		Eileen.Sideways@falconforce.traini...	Rain and Cloud specialist	Member

- Overview
- Quickstart
- Integration assistant
- Diagnose and solve problems
- Manage
 - Branding & properties
 - Authentication
 - Certificates & secrets
 - Token configuration
 - API permissions
 - Expose an API
 - App roles
 - Owners**
 - Roles and administrators
 - Manifest
- Support + Troubleshooting



ATTACK PATH ENRICHMENT ➤ FALCONHOUND

Query

Neo4j
MDE
M365 Security
MS Graph API
Azure Data Explorer
Sentinel
Splunk
Elastic
Falcon Logscale
BloodHound API

Process

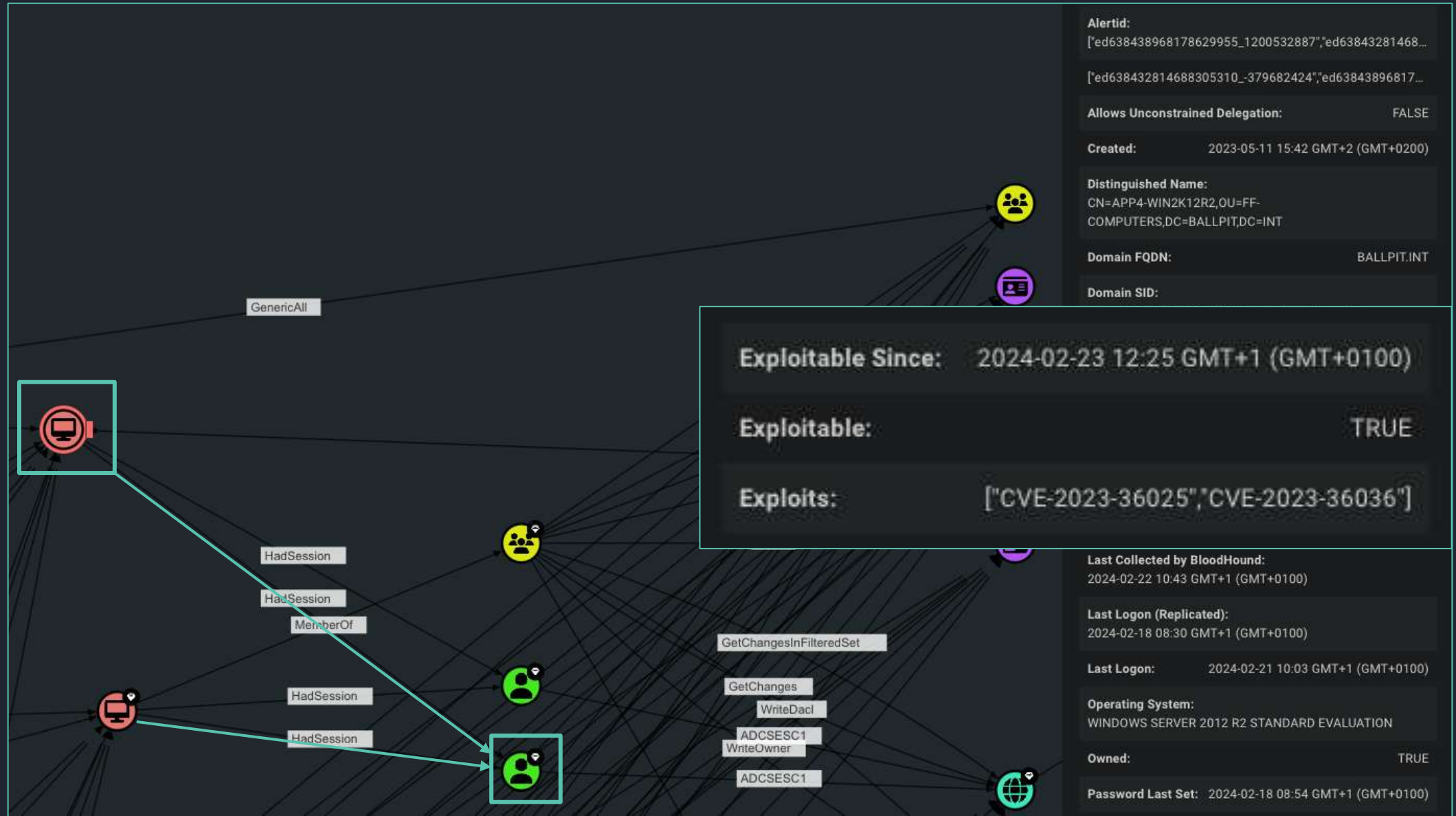
FalconHound
Actions

Store

Sentinel table
Sentinel Watchlist
Neo4j
CSV / JSON / MD
Azure Data Explorer
Splunk
Falcon Logscale
Bloodhound API
Elastic

<https://github.com/FalconForceTeam/FalconHound>

EXPLOITABLE MACHINES TO HIGH VALUE RESOURCES



WHICH GOT ME THINKING

- ☐ Kusto supports a graph now
- ☐ What if I can get the BloodHound info in Kusto, without too much hassle for the user
- ☐ Can it do the same as in Neo4j ?
- ☐ How hard can it be ?



TURNS OUT, IT WAS, A BIT



THE PLAN

Use the AzureHound scan data

Look at how BloodHound postprocesses edges

Develop an ingestor to ADX that creates the same nodes and edges

Query attack paths in Kusto

Integrate FalconHound

BLOODHOUND IS OPEN SOURCE

An example post processing function

```
func createAZMGApplicationReadWriteAllEdges(ctx context.Context, db graph.Database, operation analysis.StatTrackedOperation) error {
    if err := db.ReadTransaction(ctx, func(tx graph.Transaction) error {
        if tenantContainsAppRelationships, err := fetchTenantContainsRelationships(tx, tenant, azure.App); err != nil {
            return err
        } else if sourceNodes, err := aggregateSourceReadWriteServicePrincipals(tx, tenantContainsServicePrincipalRelationships); err != nil {
            return err
        } else {
            return operation.Operation.SubmitReader(func(ctx context.Context, tx graph.Transaction, outC chan<- analysis.Result) error {
                for _, targetRelationship := range append(tenantContainsServicePrincipalRelationships, tenantContainsAppRelationships) {
                    for _, sourceNode := range sourceNodes {
                        AZMGAddSecretRelationship := analysis.CreatePostRelationshipJob{
                            FromID: sourceNode.ID,
                            ToID:   targetRelationship.EndID,
                            Kind:   azure.AZMGAddSecret,
                        }

                        if !channels.Submit(ctx, outC, AZMGAddSecretRelationship) {
                            return nil
                        }

                        AZMGAddOwnerRelationship := analysis.CreatePostRelationshipJob{
                            FromID: sourceNode.ID,
                            ToID:   targetRelationship.EndID,
                            Kind:   azure.AZMGAddOwner,
                        }

                        if !channels.Submit(ctx, outC, AZMGAddOwnerRelationship) {
                            return nil
                        }
                    }
                }
            })
        }
    })
    return nil
}
```

Several post processed edges to be added

```
943 PathfindingRelationships: list.Concat([InboundOutboundRelationships,
944
945 PostProcessedRelationships: [
946     AddSecret,
947     ExecuteCommand,
948     ResetPassword,
949     AddMembers,
950     GlobalAdmin,
951     PrivilegedRoleAdmin,
952     PrivilegedAuthAdmin,
953     AZMGAddMember,
954     AZMGAddOwner,
955     AZMGAddSecret,
956     AZMGGrantAppRoles,
957     AZMGGrantRole,
958     SyncedToADUser,
959     AZRoleApprover,
960 ]
```


CUSTOM BLOODHOUND INGEST

Designed for ingestion in Azure Data Explorer since this has support for

- Kusto Graph Semantics
- GraphQL
- OpenCypher

Also, graph snapshots which allow for faster searching and historical comparison.

Additional benefits;

Speed

Updateable rows

Non-timeseries option

TRANSFORMING SCANS



sharphound.zip

Use the SharpHound scan data

Use the AzureHound scan data

Look at how
BloodHound
postprocesses
edges



Develop an ingestor to
ADX that creates the
same nodes and edges

Nodes table

Edges table

create a
graph model



azurehound.json

```
134 AzureEdges
135 | sample 10
```

Table 1														+ Add visual	Stats	Search	2025-11-04 07:58 (UTC)	Done (0.124 s)	# 10 records	Hide empty columns		
id	sourceId	targetId	type	edgeType	label	weight	properties	timestamp														
> 19536a84-0da6-4cc5-9ba7-	19536a84-0da6-4cc5-9ba7-	2d947f80-7b1c-4bf5-b1	AZMGAddSecret	AZMGAddSecret	Can Add Secret	10	{"description":"Can add secrets to this object	10/22/2025, 5:44:47.933 PM														
> 19536a84-0da6-4cc5-9ba7-	19536a84-0da6-4cc5-9ba7-	19369dd0-77f2-4b2c-9	AZMGAddSecret	AZMGAddSecret	Can Add Secret	10	{"description":"Can add secrets to this object	10/22/2025, 5:44:47.933 PM														
> 1c1cd450-b0e4-486b-a9d4-	1c1cd450-b0e4-486b-a9d4-	47c2692a-5be7-4d1f-9f	AZMGAddSecret	AZMGAddSecret	Can Add Secret	10	{"description":"Can add secrets to this object	10/22/2025, 5:44:47.933 PM														
> 1c1cd450-b0e4-486b-a9d4-	1c1cd450-b0e4-486b-a9d4-	8c66553c-0567-48ab-4	AZMGAddSecret	AZMGAddSecret	Can Add Secret	10	{"description":"Can add secrets to this object	10/22/2025, 5:44:47.933 PM														
> e8f93197-a229-4733-bff4-1-	e8f93197-a229-4733-b	84234d60-576f-488f-a	AZMGAddSecret	AZMGAddSecret	Can Add Secret	10	{"description":"Can add secrets to this object	10/22/2025, 5:44:47.933 PM														
> FALCONFORCE.LOCAL-S-1-5	FALCONFORCE.LOCAL-S-1-5	CEAB8E1F-D562-4A95-	WriteDacl	WriteDacl	WriteDacl	9	{"inheritanceHash":"","isInherited":false,"princ	11/1/2025, 1:21:58.263 PM														
> S-1-5-21-1629635520-12961	S-1-5-21-1629635520-12961	913AF001-C473-4F21-E	WriteDacl	WriteDacl	WriteDacl	9	{"inheritanceHash":"","isInherited":false,"princ	11/1/2025, 1:21:58.263 PM														
> FALCONFORCE.LOCAL-S-1-5	FALCONFORCE.LOCAL-S-1-5	19F949B5-56BA-49A4-	WriteDacl	WriteDacl	WriteDacl	9	{"inheritanceHash":"","isInherited":false,"princ	11/1/2025, 1:21:58.263 PM														
> S-1-5-21-1629635520-12961	S-1-5-21-1629635520-12961	S-1-5-21-1629635520-	GenericAll	GenericAll	GenericAll	10	{"inheritanceHash":"","isInherited":false,"princ	11/1/2025, 1:21:58.263 PM														
> FALCONFORCE.LOCAL-S-1-5	FALCONFORCE.LOCAL-S-1-5	S-1-5-21-1629635520-	AllExtendedRights	AllExtendedRights	AllExtendedRights	8	{"inheritanceHash":"","isInherited":false,"princ	11/1/2025, 1:21:58.263 PM														

```
134 AzureNodes
135 | sample 10
```

Table 1				+ Add visual	Stats	Search	2025-11-04 07:59 (UTC)	Done (0.070 s)	# 10 records	Hide empty columns		
id	type	label	properties	tenantid	timestamp							
> 58dacdf4-67be-436c-9912-3ba52191c887	AZServicePrincipal	Azure Monitor for VMs	{"displayName":"Azure Monitor for VMs","id":"58dacdf4-67be-436c-9912-3ba52191c887","tenantId":"960b4203-960b4203-ca5f-4c64"	960b4203-ca5f-4c64	10/22/2025, 5:44:46.51 PM							
> 966707d0-3269-4727	AZRole	Password Administrator	{"description":"Can reset passwords for non-administrators and Password Administrators.","displayName":"Passwo	960b4203-ca5f-4c64	10/22/2025, 5:44:46.51 PM							
> S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
> 71F91FA1-DCBC-42C2	Container	Container	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
> S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
> S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
> S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							
> FALCONFORCE.LOCAL	Group	Group	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"FALCONFORCE.LOCAL-S-1-5-32-544","PrincipalI		11/1/2025, 1:21:54.225 PM							
> S-1-5-21-162963552C	User	User	{"Aces":{"InheritanceHash":"","IsInherited":false,"PrincipalSID":"S-1-5-21-1629635520-1296687596-174493759-		11/1/2025, 1:21:54.225 PM							

GRAPH CREATION OPTIONS

make-graph

```
let Nodes = AzureNodes
  | extend properties=tostring(properties)
  | project id, type, name=label, tenantId, properties;
let Edges = AzureEdges
  | extend properties=tostring(properties)
  | project sourceId, targetId, type, properties,
name=label, label=edgeType;
Edges
  | make-graph sourceId --> targetId with Nodes on id
```

.create-or-alter
graph_model

GRAPH CREATION OPTIONS

make-graph

.create-or-alter
graph_model

```
.create-or-alter graph_model AzureHound ```
{"Definition": {
  "Steps": [
    {
      "Kind": "AddNodes",
      "Query": "AzureNodes",
      "NodeIdColumn": "id",
      "Labels": ["AZBase"],
      "LabelsColumn": "type"
    },
    {
      "Kind": "AddEdges",
      "Query": "AzureEdges",
      "SourceColumn": "sourceId",
      "TargetColumn": "targetId",
      "LabelsColumn": "edgeType"
    }
  ]
}
}```
```

BASIC KQL GRAPH EXAMPLE

Table 1

+

Add visual

Stats

Search

from

via

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

JPath:

/from

Inline

Full

```

1  "from": Eileen Sideways,
2  "via": [
3    {
4      "sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
5      "targetId": "a418e18a-592b-46f6-ad20-95b004355529",
6      "type": "Owns",
7      "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"},
8      "name": "Owns App",
9      "label": "AppOwner"
10   },
11  ]

```

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

Eileen Sideways

[{"sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb", "targetId": "a418e18a-592b-46f6-ad20-95b004355529", "type": "Owns", "properties": {"relationship": "AppOwner", "targetName": "Weather prediction app (Demo)", "targetType": "AZApp", "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"}, "name": "Owns App", "label": "AppOwner"}]

KQL GRAPH EXAMPLE

```
198  AttackGraph
199  | graph-match (u)-[e*1..4]->(d)
200  |   where u.type contains "AZUser"
201  |   and d.name contains "Global Administrator"
202  |   project from = u.name, via = e, to = d.name
203  | where from contains "Eileen"
204  | take 1
205  | mv-expand hop = via to typeof(dynamic)
206  | where isnotnull(hop)
207  | extend rel = tostring(hop.label),
208  |   | prop = parse_json(tostring(hop.properties))
209  | extend tgtName = tostring(prop.targetName)
210  | serialize
211  | extend hop_index = row_number(0)
212  | extend srcName = iif(hop_index == 0, from, prev(tgtName))
213  | project path_segment = strcat(srcName, " -[", rel, "]"-> ", tgtName)
214
215
```

 Table 1  Add visual  Stats

path_segment

- > Eileen Sideways —[AppOwner]→ Weather prediction app (Demo)
- > Weather prediction app (Demo) —[RunAs]→ Weather prediction app (Demo)
- > Weather prediction app (Demo) —[AZMGAddOwner]→ Windows Azure Service Management API
- > Windows Azure Service Management API —[AZMGGrantRole]→ Global Administrator

OPEN CYPHER SUPPORT!

```
#crp_query_graph_reference = graph('AzureHound')
#crp_query_language = opencypher
MATCH p=(u:AZUser)-[e*1..4]->(z:AZRole) WHERE u.properties.displayName CONTAINS "Eileen"
and z.properties.displayName CONTAINS "Global Administrator" RETURN p

P
[
  {
    "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
    "type": "AZUser",
    "label": "Eileen Sideways",
    "properties": {
      "accountEnabled": true,
      "displayName": "Eileen Sideways",
    },
  },
  {
    "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
    "type": "AZUser",
    "label": "Eileen Sideways",
    "properties": {
      "accountEnabled": true,
      "displayName": "Eileen Sideways",
    },
  },
  {
    "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
    "type": "AZUser",
    "label": "Eileen Sideways",
    "properties": {
      "accountEnabled": true,
      "displayName": "Eileen Sideways",
    },
  },
  {
    "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
    "type": "AZUser",
    "label": "Eileen Sideways",
    "properties": {
      "accountEnabled": true,
      "displayName": "Eileen Sideways",
    },
  },
]
Done (00:00.304): 4 records
```

```
{
  "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
  "type": "AZUser",
  "label": "Eileen Sideways",
  "properties": {
    "accountEnabled": true,
    "displayName": "Eileen Sideways",
    "id": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
    "tenantId": "960b4203-ca5f-4c64-a9c1-a690bc0112e5",
    "tenantName": "FalconForce Training",
    "createdDateTime": "2023-06-08T14:19:09.0000000Z",
    "jobTitle": "Rain and Cloud specialist",
    "lastPasswordChangeDateTime": "2023-06-08T14:19:09.0000000Z",
    "userPrincipalName": "Eileen.Sideways@falconforce.training",
    "userType": "Member"
  },
  "tenantId": "960b4203-ca5f-4c64-a9c1-a690bc0112e5",
  "timestamp": "2025-08-18T11:10:14.5230000Z",
  "$labels": [
    "AZBase",
    "AZUser"
  ]
},
{
  "id": "a418e18a-592b-46f6-ad20-95b004355529_owner_41e417c0-4fba-47a3-8511-b4a59e2a14eb_0",
  "sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
  "targetId": "a418e18a-592b-46f6-ad20-95b004355529",
  "type": "Owns",
  "edgeType": "AppOwner",
  "label": "Owns App",
  "weight": 10.0,
  "properties": {
    "relationship": "AppOwner",
    "targetName": "Weather prediction app (Demo)",
    "targetType": "AZApp",
    "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"
  },
  "timestamp": "2025-08-18T11:10:16.4720000Z",
  "$labels": [
    "AppOwner"
  ]
},
}
```

EDGE WEIGHT EXPERIMENT

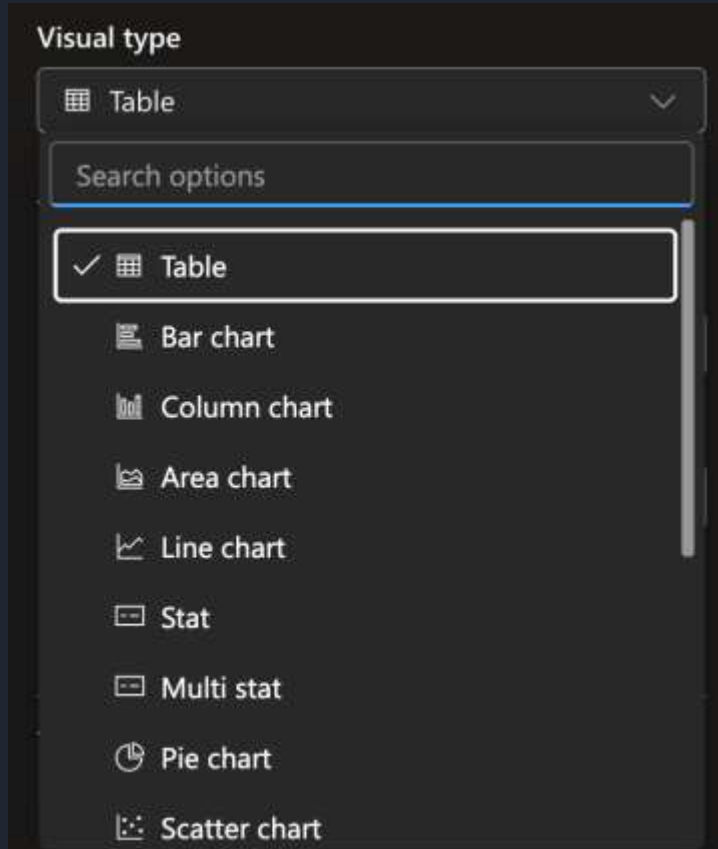
Trying out assigning an ease-of-exploitation weight .

This allows path for example to show the path of least resistance or most likely to be abused one.

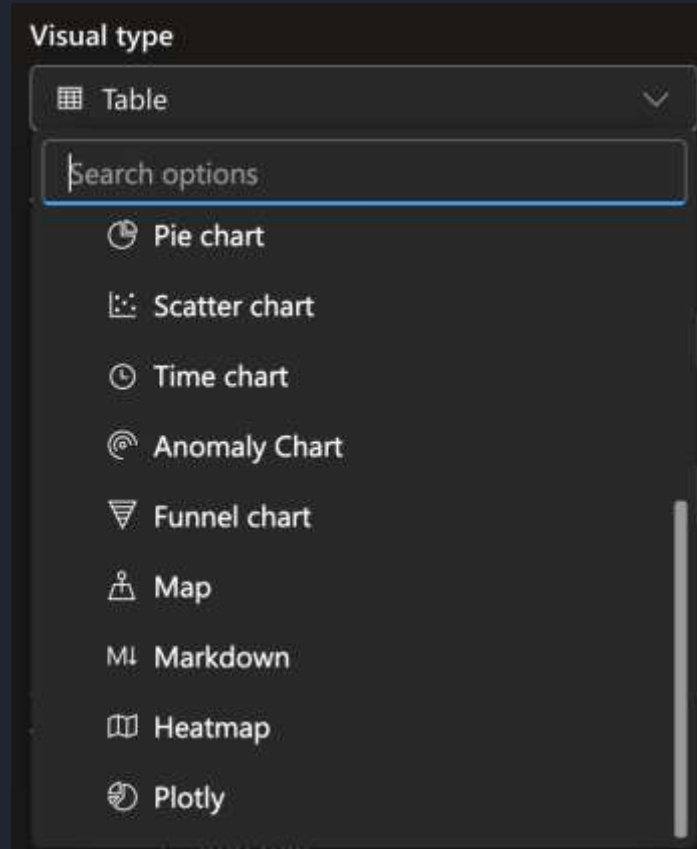
Granted, there's may be a degree of arbitrariness here.

```
{
  "id": "a418e18a-592b-46f6-ad20-95b004355529_owner_41e417c0-4fba-47a3-8511-b4a59e2a14eb_0",
  "sourceId": "41e417c0-4fba-47a3-8511-b4a59e2a14eb",
  "targetId": "a418e18a-592b-46f6-ad20-95b004355529",
  "type": "Owns",
  "edgeType": "AppOwner",
  "label": "Owns App",
  "weight": 10.0,
  "properties": {
    "relationship": "AppOwner",
    "targetName": "Weather prediction app (Demo)",
    "targetType": "AZApp",
    "appId": "e124f9f7-597c-4f0e-8311-ad71e4fd4d8a"
  },
  "timestamp": "2025-08-18T11:10:16.4720000Z",
  "$labels": [
    "AppOwner"
  ]
}
```

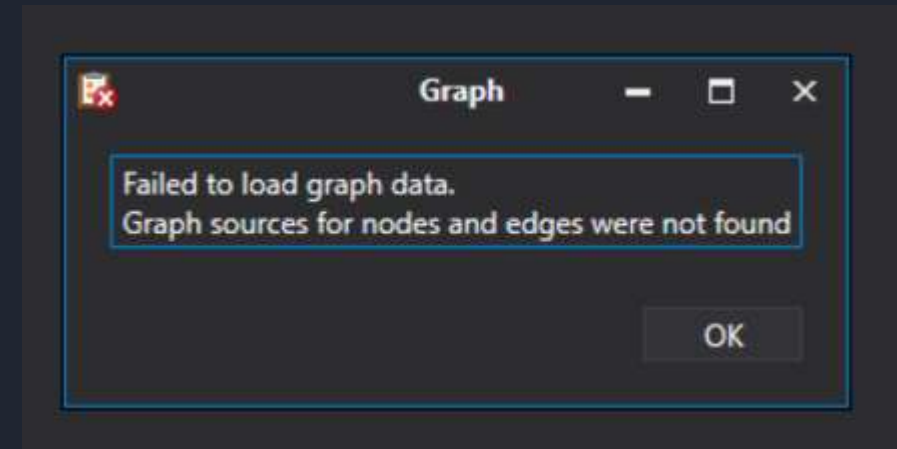
NO VISUAL GRAPH SUPPORT 😞



Azure Data Explorer



Kusto Explorer



GRAPH VISUALIZATION OVERVIEW

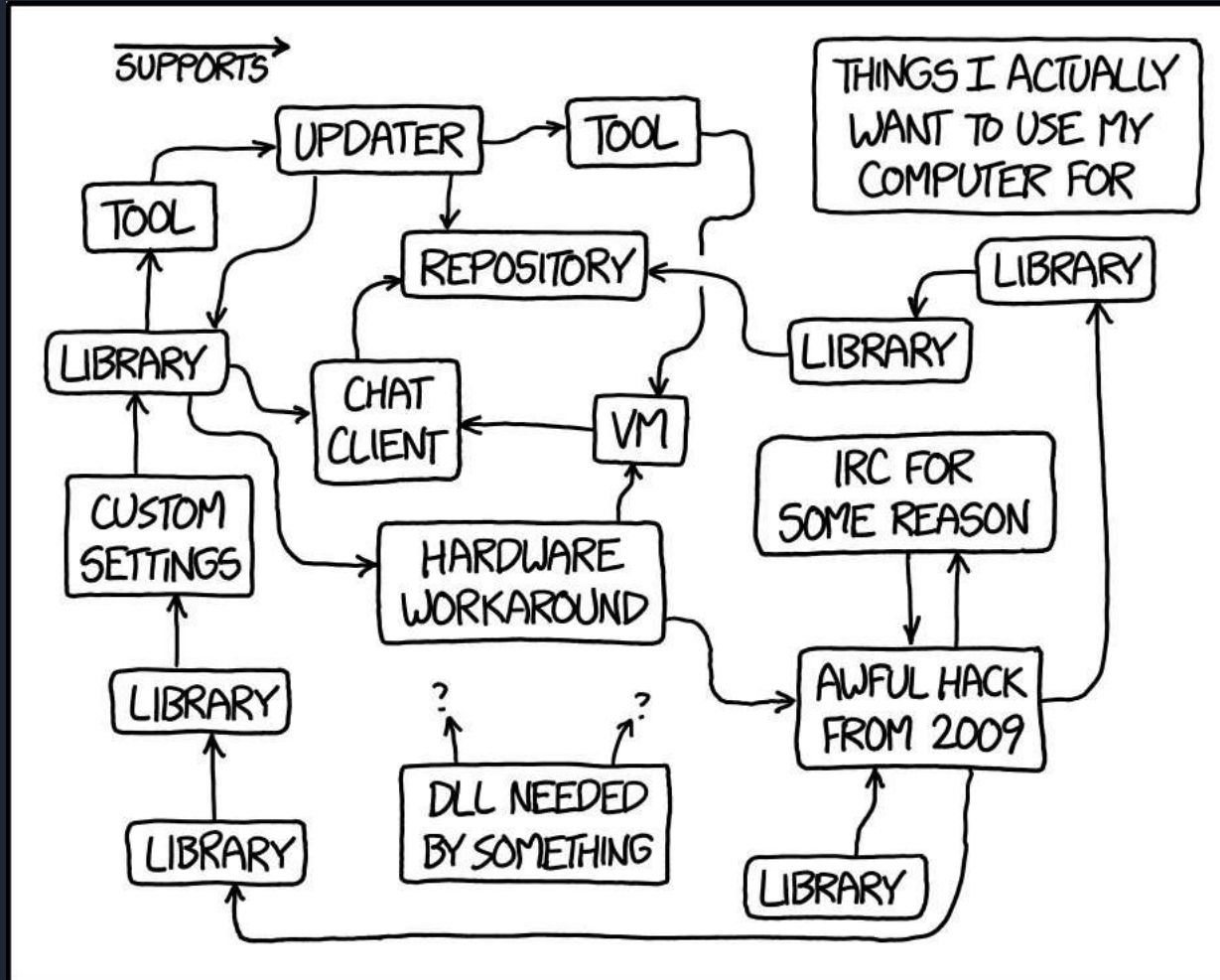


Kusto Explorer

Kusto Explorer provides built-in graph visualization capabilities that automatically render interactive graph visualizations when your KQL queries end with the `make-graph` operator or use the `graph()` function.

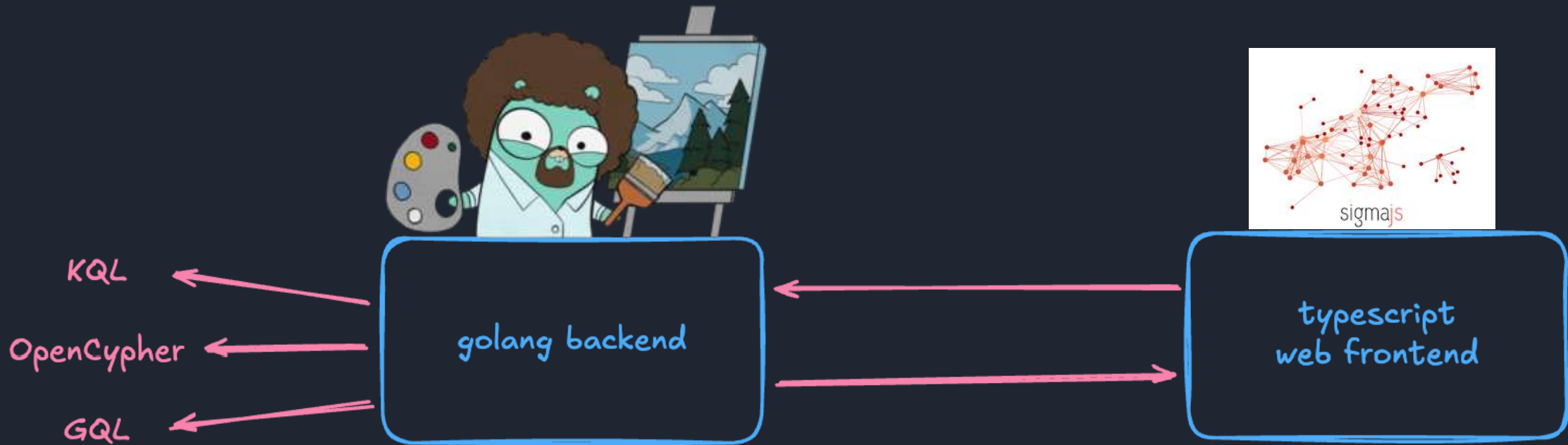
Graphistry

Graphistry offers GPU-accelerated graph visualization and analytics that integrates with KQL graph semantics. This platform enables interactive exploration of large-scale graph datasets with advanced visual analytics capabilities.



EVERY NOW AND THEN I REALIZE I'M MAINTAINING A HUGE CHAIN OF TECHNOLOGY SOLELY TO SUPPORT ITSELF.

'SIMPLE' INTERFACE



SET ffse-fa-braininterface-b5a24a.a...

POST http://localhost:8080/api/query

JSON Params Headers Auth Info

```
1 {
2   "language": "kql",
3   "query": "graph('AzureHound',true) | graph-match (u)-[e]->(d)
4   where e contains \"MemberOf\" project from=u, via=e, to=d"
```

Calypso

POST KQL graph 200
POST Cypher graph 200
POST Cypher graph 200

KQL graph

200 OK • 406 ms • 849.3 KB

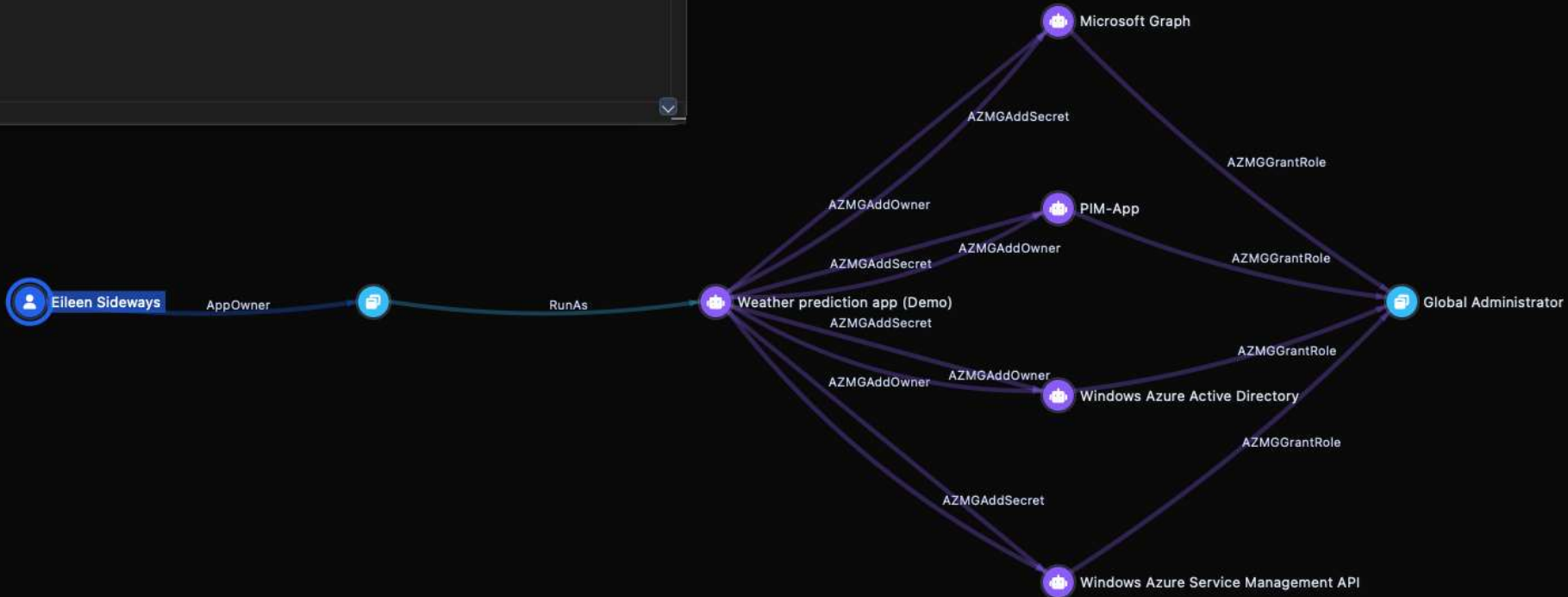
Pretty Headers Info

```
1 {
2   "columns": [
3     {
4       "field": "from",
5       "title": "from",
6       "type": "dynamic"
7     },
8     {
9       "field": "via",
10      "title": "via",
11      "type": "dynamic"
12     },
13     {
14       "field": "to",
15       "title": "to",
16       "type": "dynamic"
17     }
18   ],
19   "rows": [
20     {
21       "from": {
22         "id": "db928c8f-519e-4748-a324-37192d4ed44c",
23         "label": "Gary Groove",
24         "properties": {
25           "accountEnabled": true,
26           "createdDateTime": "2024-07-25T14:33:00.0000000Z",
27           "displayName": "Gary Groove",
28           "id": "db928c8f-519e-4748-a324-37192d4ed44c",
29           "jobTitle": "Piano Tuner",
30           "lastPasswordChangeDateTime": "2024-07-26T09:24:23.0000000Z",
31           "tenantId": "960b4203-ca5f-4c64-a9c1-a690bc0112e5",
32           "tenantName": "FalconForce Training",
33           "userPrincipalName": "gary.groove@falconforce.training",
34           "userType": "Member"
35         },
36         "tenantId": "960b4203-ca5f-4c64-a9c1-a690bc0112e5",
37         "timestamp": "2025-10-22T17:44:46.5100000Z",
38         "type": "AZUser"
39       },
40       "to": {
41         "id": "1a161e7f-5a80-4fd5-83d6-c36e403eb05a",
42         "label": "All Users",
43         "properties": {
44           "createdDateTime": "2024-07-24T05:56:59.0000000Z",
45           "displayName": "All Users",
46           "id": "1a161e7f-5a80-4fd5-83d6-c36e403eb05a",
47           "mailNickname": "d22f9195-f7a3-4f43-8e1c-fe686856442f",
48           "securityEnabled": true,
49           "securityIdentifier": "S-1-12-1-437657215-1339382400-1858328195-1521499712",
50           "tenantId": "960b4203-ca5f-4c64-a9c1-a690bc0112e5",
```

PREVIEWING CALYPSO

KQL Cypher GQL Radial Layered Force Grid Light Mode Hide Labels Hide Editor Hide Details Show Results

```
MATCH p=(u:AZUser)-[*1..4]->(z:AZRole) WHERE u.properties.displayName CONTAINS "Eileen" and z.properties.displayName CONTAINS "Global Administrator" RETURN p
```



Node Details

LABEL

 Eileen Sideways

TYPE

AZUser

IDENTIFIER

41e417c0-4fba-47a3-8511-b4a59e2a14eb

ALL PROPERTIES

Labels

["AZBase", "AZUser"]

accountEnabled

true

createdDateTime

2023-06-08T14:19:09.000000Z

displayName

Eileen Sideways

id

41e417c0-4fba-47a3-8511-b4a59e2a14eb

jobTitle

Rain and Cloud specialist

lastPasswordChangeDateTime

2023-06-08T14:19:09.000000Z

tenantId

960b4203-ca5f-4c64-a9c1-a690bc0112e5

tenantName

FalconForce Training

timestamp

2025-10-22T17:44:46.510000Z

userPrincipalName

Eileen.Sideways@falconforce.training

userType

INCLUDING KQL SUPPORT

KQLCypherGQLRadialLayeredForceGridLight ModeHide LabelsHide EditorHide DetailsHide Results

graph('AzureHound')
| graph-match (u)-[e*1..4]->(d) where u contains "AZVM" and d contains
"User Administrator" project frommu, viae, to=d

Run

DC-1

Server_ManagedIdentity

User Administrator

AZManagedIdentity

RoleAssignment

Node Details

LABEL
DC-1

TYPE
AZVM

IDENTIFIER
/subscriptions/19b31a7d-b2fc-4e70-9cb4-f4ec5ccbbb0d/resourceGroups/STUDENT-RG-MOONLIT-1/providers/Microsoft.Compute/virtualMachines/DC-1

ALL PROPERTIES

id
/subscriptions/19b31a7d-b2fc-4e70-9cb4-f4ec5ccbbb0d/resourceGroups/STUDENT-RG-MOONLIT-1/providers/Microsoft.Compute/virtualMachines/DC-1

identity
{ "principalId": "", "tenantId": "", "type": "UserAssigned", "userAssignedIdentities": { "/subscriptions/19b31a7d-b2fc-4e70-9cb4-f4ec5ccbbb0d/resourceGroups/STUDENT-RG-MOONLIT-1/providers/Microsoft.ManagedIdentity/userAssignedIdentities/Server_ManagedIdentity": { "clientId": "bcea4502-66ea-42d2-b63e-cb8c448614c6", "principalId": "74a05c2e-b384-49ac-80e5-28a76242bcfd" } } }

location
westeurope

name
DC-1

properties
{ "additionalCapabilities": {}, "applicationProfile": {}, "availabilitySet": {}, "billingProfile": {}, "capacityReservation": { "capacityReservationGroup": {} }, "diagnosticsProfile": { "bootDiagnostics": {}, "extensionsTimeBudget": "PT1H30M", "hardwareProfile": { "vmSize": "Standard_B2s", "vmSizeProperties": {} }, "host": {}, "hostGroup": {}, "instanceView": { "bootDiagnostics": { "status": {} }, "maintenanceRedeployStatus": {}, "patchStatus": { "availablePatchSummary": { "error": { "code": "", "message": "" }, "lastPatchInstallationSummary": { "error": { "code": "", "message": "" } } }

BLAST RADIUS BASED ON EDGE WEIGHTS

```
282 let aggregatedPaths = (  
283   paths  
284   | sort by sourceId, targetWeight desc  
285   | summarize blastRadiusList = array_slice(make_set_if(targetId, isnotempty(targetId)), 0, (listedIdsLimit - 1))  
286     , blastRadiusScore = dcountif(targetId, isnotempty(targetId))  
287     , blastRadiusScoreWeighted = sum(targetWeight)  
288   by sourceId  
289   | extend isBlastRadiusListCapped = (blastRadiusScore > listedIdsLimit)  
290 );  
291 aggregatedPaths  
292 | top resultCountLimit by blastRadiusScore desc  
293 );  
294 graph('AzureHound')  
295 | graph-match (u)-[e*1..4]->(d) project SourceNodeName=u.id,TargetNodeName=d.id,TargetNodeCriticality=e.weight  
296 | invoke graph_blast_radius_fl(sourceIdColumnName = 'SourceNodeName'  
297   , targetIdColumnName = 'TargetNodeName'  
298   , targetWeightColumnName = 'TargetNodeCriticality'  
299 )  
300
```

Table 1 + Add visual Stats

Search

2025-11-06 13:10 (UTC)

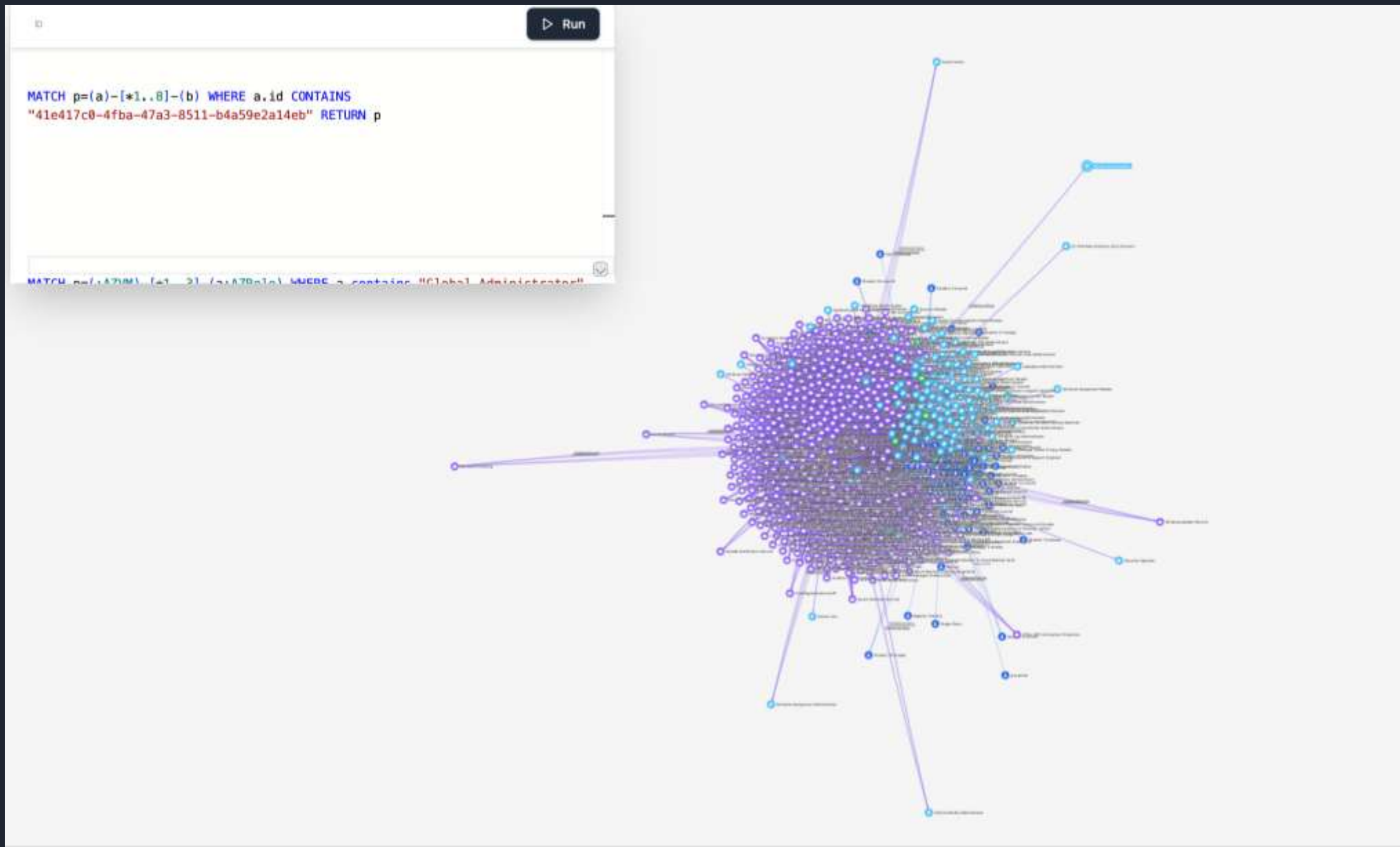
Done (77.986 s)

206 records

Hide

sourceId	blastRadiusList	blastRadiusScore
> S-1-5-21-1629635520-1296687596-174493759-512	["S-1-5-21-1629635520-1296687596-174493759-1186","S-1-5-21-1629635520-1296687596-174493759-1674","S-1-5-21-1629635520-1296687596-174493759-513","S-1-5-21-1629635520-1296687596-174493759-512"]	1,382
> S-1-5-21-1629635520-1296687596-174493759-500	["S-1-5-21-1629635520-1296687596-174493759-520","S-1-5-21-1629635520-1296687596-174493759-519","S-1-5-21-1629635520-1296687596-174493759-518","S-1-5-21-1629635520-1296687596-174493759-500"]	1,382
> S-1-5-21-1629635520-1296687596-174493759-519	["S-1-5-21-1629635520-1296687596-174493759-1471","EA74A08D-E040-47CF-9D3F-B63E7D27ED23","S-1-5-21-1629635520-1296687596-174493759-1277","S-1-5-21-1629635520-1296687596-174493759-519"]	1,382
> FALCONFORCE.LOCAL-S-1-5-32-544	["S-1-5-21-1629635520-1296687596-174493759-1773","385FAFC0-F25A-4CA0-80E5-B28198B64277","S-1-5-21-1629635520-1296687596-174493759-1432","S-1-5-21-1629635520-1296687596-174493759-544"]	1,382
> FALCONFORCE.LOCAL-S-1-5-32-548	["S-1-5-21-1629635520-1296687596-174493759-1980","S-1-5-21-1629635520-1296687596-174493759-1657","S-1-5-21-1629635520-1296687596-174493759-1170","S-1-5-21-1629635520-1296687596-174493759-548"]	1,093
> S-1-5-21-1629635520-1296687596-174493759-527	["S-1-5-21-1629635520-1296687596-174493759-1622","S-1-5-21-1629635520-1296687596-174493759-2088","S-1-5-21-1629635520-1296687596-174493759-2078","S-1-5-21-1629635520-1296687596-174493759-527"]	1,042
> S-1-5-21-1629635520-1296687596-174493759-526	["S-1-5-21-1629635520-1296687596-174493759-1694","S-1-5-21-1629635520-1296687596-174493759-1673","S-1-5-21-1629635520-1296687596-174493759-1132","S-1-5-21-1629635520-1296687596-174493759-526"]	1,042
> 41e417c0-4fba-47a3-8511-b4a59e2a14eb	["547501a6-d166-48ac-9f81-25df02e7d4af","1a161e7f-5a80-4fd5-83d6-c36e403eb05a","a418e18a-592b-46f6-ad20-95b004355529","d3e3eb0d-2686-4738-b5c2-13b42580fbf0","bce9363e-cfae-4534-87f3-ecb7199ae08f","620c8788-709b-4a7e-8968-02ab5f93421f","8a2253ad-a459-4e60-bd14-cc19536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> 1c1cd450-b0e4-486b-a9d4-c512ec9f5aa2	["ec59d28e-7dda-4480-8110-b1bb792e5604","3169fe2c-2086-42d3-ab1f-2baf85f25b59","ec4b4808-1b81-4dee-89af-f7b06e1bc60c","16a0103a-e9db-426f-996f-8342-d9273494-a75e-4840-be1e-19bc0e2a9af2","75b922e4-3be6-455c-9959-5023dd41697e","b70f1d37-f647-4940-8a0d-e4e5379440f9","8d88fa99-e1b1-4500-a524-8653c2d7dfea","5cf0e0c3-600b-422e-958d-519536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> 2e1a74be-db05-4bae-a49f-42025055c70d	["e8f93197-a229-4733-bff4-1322aa0d5a78","44c59f8b-64ad-42ff-bb98-9710e43108ea","78ed25f8-075b-4daa-a0df-da6b218ff1b2","2773a6e7-0aff-49d5-b67e-221e3b42025055c70d","75b922e4-3be6-455c-9959-5023dd41697e","b70f1d37-f647-4940-8a0d-e4e5379440f9","8d88fa99-e1b1-4500-a524-8653c2d7dfea","5cf0e0c3-600b-422e-958d-519536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> d9273494-a75e-4840-be1e-19bc0e2a9af2	["75b922e4-3be6-455c-9959-5023dd41697e","b70f1d37-f647-4940-8a0d-e4e5379440f9","8d88fa99-e1b1-4500-a524-8653c2d7dfea","5cf0e0c3-600b-422e-958d-519536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> 1e0053a9-75bc-4dd5-b973-229f5afb8f4d	["965dd7c2-fed8-49a0-94b9-618d8501be53","e2b510e2-0938-4451-9126-92321b4ef9e9","84abb822-ae45-477f-bc78-ffce680e3b6d","7c369c77-0110-402b-b7a0-77a0053a9-75bc-4dd5-b973-229f5afb8f4d","75b922e4-3be6-455c-9959-5023dd41697e","b70f1d37-f647-4940-8a0d-e4e5379440f9","8d88fa99-e1b1-4500-a524-8653c2d7dfea","5cf0e0c3-600b-422e-958d-519536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> a418e18a-592b-46f6-ad20-95b004355529	["d3e3eb0d-2686-4738-b5c2-13b42580fbf0","bce9363e-cfae-4534-87f3-ecb7199ae08f","620c8788-709b-4a7e-8968-02ab5f93421f","8a2253ad-a459-4e60-bd14-cc19536a84","0da6-4cc5-9ba7-593878df20bc"]	722
> 19536a84-0da6-4cc5-9ba7-593878df20bc	["31daba95-bf88-4cc4-84f3-c4858be819f8","29232cdf-9323-42fd-ade2-1d097af3e4de","cc0cf5f1-d9c6-4087-bdf8-b93bc4d30c6b","b0e8c612-f722-425c-bc57-eb719536a84-0da6-4cc5-9ba7-593878df20bc"]	722
> a8f02107-c232-4733-bff4-1322aa0d5a78	["b1542f8a-68d1-4c26-b1d4-f68f4751c021","52809f5c-7c232-42c5-0682-9db058b73222","178a425f8-075b-4daa-a0df-da6b218ff1b2","b1bb792e5604","3169fe2c-2086-42d3-ab1f-2baf85f25b59","ec4b4808-1b81-4dee-89af-f7b06e1bc60c","16a0103a-e9db-426f-996f-8342-d9273494-a75e-4840-be1e-19bc0e2a9af2","75b922e4-3be6-455c-9959-5023dd41697e","b70f1d37-f647-4940-8a0d-e4e5379440f9","8d88fa99-e1b1-4500-a524-8653c2d7dfea","5cf0e0c3-600b-422e-958d-519536a84","0da6-4cc5-9ba7-593878df20bc"]	722

BLAST RADIUS BASED ON EDGE WEIGHTS



BLAST RADIUS BASED ON EDGE WEIGHTS

8

Run

```
MATCH p=(a)-[*1..8]-(b) WHERE a.id CONTAINS  
"41e417c0-4fba-47a3-8511-b4a59e2a14eb" RETURN p
```

```
MATCH (a) WHERE a.id CONTAINS "41e417c0-4fba-47a3-8511-b4a59e2a14eb" RETURN  
a
```



Node Details

LABEL

 Eileen Sideways

TYPE

AZUser

IDENTIFIER

41e417c0-4fba-47a3-8511-b4a59e2a14eb

ALL PROPERTIES

accountEnabled

true

createdDateTime

2023-06-08T14:19:09.000000Z

displayName

Eileen Sideways

id

41e417c0-4fba-47a3-8511-b4a59e2a14eb

jobTitle

Rain and Cloud specialist

lastPasswordChangeDateTime

2023-06-08T14:19:09.000000Z

tenantId

960b4203-ca8f-4c64-a9c1-a690bc0112e5

tenantName

FalconForce Training

timestamp

2025-10-22T17:44:46.510000Z

userPrincipalName

Eileen.Sideways@falconforce.training

[SOME] NEXT STEPS

- Merge more data into a single graph
 - Enriched nodes
 - Risk based scoring
 - Sub-graphs of alerts, DevOps, Office and more
- Connect graphs (somehow)
- Make the tools stable and open-source them
- Update FalconHound to support this model
- Support Sentinel Data Lake / Defender XDR



ACKNOWLEDGEMENT



This would never have been possible without the amazing work by the SpecterOps team.

They make all this available open-source, allowing me to experiment with this, based on all their hard work and research.

I can't emphasize enough how important this is for the community.



THANK YOU!!

Together. Secure. Today.



olaf@falconforce.nl



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